

User Requirement Document

BiasAperture: A Diagnostic and Evaluative Framework for Auditing Demographic Bias in Facial Analysis Systems

Project Title

BiasAperture: A Diagnostic and Evaluative Framework for Auditing Demographic Bias in Facial Analysis Systems

Project Context

Project Summary

BiasAperture is a diagnostic and evaluative software platform that computes subgroup and intersectional fairness metrics for a third-party facial-analysis model and reports them in a standardized, regulator-legible format. It is organized into five cooperating modules including data ingestion, model interfacing, fairness-metric computation, explainability, and report generation. Its analytical core computes four disparity metrics (demographic parity difference, equalized odds difference, equal opportunity difference, and disparate impact ratio) using AIF360 and Fairlearn as independent, cross-validating backends, with every reported disparity accompanied by a chi-squared significance test and a bootstrap confidence interval. A SHAP-based explainability layer attributes flagged disparities to input features. Findings are assembled into an exportable, Model-Cards- and Datasheets-structured HTML report in which every metric is traced to its basis under Article 10 of the EU AI Act and the corresponding function of the NIST AI Risk Management Framework. The platform is validated against the FairFace benchmark dataset. BiasAperture is scoped strictly as diagnostic it identifies and statistically characterizes disparities but does not mitigate bias, retrain models, or generate synthetic demographic data.

Problem Statement

There is a clear engineering gap between a mature and growing body of fairness metrics, benchmark datasets, and bias-mitigation research, and the availability of an accessible, reusable, standards-aligned software platform that operationalizes this research into a repeatable auditing workflow for facial analysis systems specifically. Existing audits (e.g. the Gender Shades study) are typically conducted externally, manually, and for a single point in time, and are not designed as reusable, repeatable software artefacts. This leaves model owners, auditors, and researchers without a systematic way to evaluate a facial-analysis model's performance across demographic subgroups using standardized datasets, statistically grounded fairness metrics, and a transparent, shareable report, at precisely the moment regulatory frameworks such as the EU AI Act begin to require exactly this kind of evidence.

Project Goals

- Design a modular software architecture that accepts a trained facial analysis model, or a file of its precomputed predictions, together with a demographically labelled benchmark dataset, and computes subgroup and intersectional fairness metrics from it.

- Integrate established open-source fairness toolkits (AIF360 and Fairlearn) alongside standard statistical significance testing, so every reported disparity is methodologically defensible rather than an artefact of sampling noise.
- Generate a standardized, exportable fairness report which is structurally informed by Model Cards and Datasheets for Datasets that communicates subgroup performance disparities to both technical and non-technical stakeholders.
- Validate the platform's design against publicly available, demographically annotated benchmark dataset, principally FairFace.
- Map every computed fairness metric to a specific obligation under Article 10 and Annex IV of the EU AI Act and to the corresponding function of the NIST AI RMF, so the platform's output is directly usable as compliance evidence.

Application Areas

BiasAperture is intended for independent, reproducible fairness auditing of automated facial analysis systems (perceived gender classification, age estimation, race/ethnicity prediction) deployed in contexts such as photo tagging, demographic analytics, surveillance, and access control. Primary users include model owners and developers auditing their own systems, independent researchers and auditors evaluating third-party models, and compliance/risk teams that need regulator-legible evidence of bias assessment under frameworks such as the EU AI Act and the NIST AI Risk Management Framework.

Project Justification

Automated facial analysis systems are widely deployed despite well-documented accuracy disparities across demographic subgroups (e.g. the Gender Shades findings of error rates as high as 34.7% for the worst-performing intersectional subgroup versus 0.8% for the best-performing one). Independent, systematic, reproducible auditing of these systems remains rare in practice, and most existing audits are manual, ad hoc, and non-reproducible. The EU AI Act (Regulation 2024/1689) imposes binding data-governance, representativeness, and bias-detection obligations on high-risk AI systems including biometric categorization under Article 10, entering into force 2 August 2026. Organizations that have not embedded documentation and measurement infrastructure ahead of this deadline face substantially higher retrofit costs. BiasAperture closes the gap between the mature fairness-metrics research literature and the absence of a reusable, standards-aligned auditing artefact, benefiting model owners, auditors, researchers, and regulators alike.

Limitations & Constraints

- BiasAperture does not mitigate bias, retrain or fine-tune models, or generate synthetic demographic data its function is strictly diagnostic, not corrective.

Any subgroup below a minimum sample size of $n \geq 30$ is flagged as statistically insufficient rather than scored.

- Full-dataset evaluation is bounded by a documented runtime target (≤ 4 hours on a single mid-range GPU for FairFace's 97,698 images) rather than guaranteed instantaneous.
- No specialized or custom hardware is required, but a mid-range GPU (≥ 8 GB VRAM) is recommended for full-dataset runs; the platform performs inference and metric computation only, never model training from scratch.

Assumptions

- The auditor supplies either a working PyTorch/TensorFlow model exposing a standard predict interface, or a precomputed predictions file in CSV/JSON format.
- Benchmark dataset (FairFace) or custom datasets supplied by the user carry usable demographic annotation fields (race/ethnicity, perceived gender, age group).
- Users have access to at minimum a 64-bit CPU environment (4+ cores, 8GB RAM) for development-scale runs, and optionally a mid-range GPU (e.g. via Google Colab) for full-dataset runs.
- Disparity thresholds (e.g. four-fifths-rule style cut-offs) are not treated as pass/fail criteria; users interpreting the report are expected to consider statistical significance and confidence intervals rather than a bare threshold.
- Users acknowledge and comply with the licensing terms of any benchmark or custom dataset before running an audit.

Data Governance & Privacy

- All input datasets and model artifacts used in an audit shall be accompanied by provenance metadata: source URL, download checksum (SHA-256), license text or URL, owner/contact, and any consent or deletion constraints supplied by the dataset owner.
- The platform shall log and embed a record of the user's explicit dataset-licensing acknowledgement into the evidence manifest (see FR-010).
- The platform shall provide a local-only processing mode in which files never leave the user's execution environment (no uploads to remote servers), and a cloud mode requiring explicit consent and encryption-in-transit.
- The platform shall offer configurable retention-policy settings (default: no retention beyond the audit unless the user explicitly requests storage) and role-based access controls for stored report bundles.
- If a dataset is de-listed or deprecated, the rationale, the commit SHA/PR removing it, and the date shall be recorded in the manifest and the project changelog.

Model Handling Modes

- Local-only mode: the model and data are processed entirely locally, with no uploads; recommended for sensitive or biometric data.
- Cloud-assisted mode: users may upload predictions or model artifacts with explicit consent; all uploads are encrypted in transit, and users may choose to store or delete cloud copies.
- A --local-only CLI flag enforces local processing; the cloud-upload workflow requires explicit, clearly labelled consent before any data leaves the user's environment.

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Requirements

Expected Inputs

- A benchmark dataset: FairFace dataset with demographic annotation fields (race/ethnicity, perceived gender, age group) and images.
- A target facial-analysis model: either a PyTorch/TensorFlow model object for direct in-process inference, or a precomputed predictions file (CSV or JSON).
- Audit configuration: selected dataset, model/predictions source, and the fairness metrics to compute, supplied via a command-line configuration interface.
- Explicit licensing acknowledgement for the selected dataset prior to execution.

Expected Outputs

- Per-subgroup and intersectional performance metrics (accuracy, precision, recall, F1, false-positive/false-negative rates).
- The four Core Four disparity metrics: DPD, EOD, EOP, DIR each with a chi-squared p-value and a 95% bootstrap confidence interval, or an insufficient sample flag.
- SHAP-based feature-attribution output for every flagged disparity.
- A standardized, exportable HTML fairness report structured on the Model Cards and Datasheets for Datasets conventions, with every metric traced to its EU AI Act Article 10/Annex IV clause and NIST AI RMF function.

Supported Task Types

- Binary classification (default): the Core Four metrics — DPD, EOD, EOP, DIR — apply directly.
- Multiclass classification: per-class metrics and one-vs-rest parity checks are computed, with per-class confusion matrices included in the report.
- Regression (e.g. continuous age estimation): subgroup RMSE/MAE and discretized-bin parity checks are computed, with subgroup calibration plots reported where applicable.
- The report clearly labels the detected task type and automatically selects the appropriate metric panel for it.

End-User & Stakeholder Requirements

Requirement (User Story)	Details	Priority	Success Criteria
As an auditor, I want to load a benchmark or custom demographically labelled dataset, so that I can evaluate a facial-analysis model's fairness on it.	Dataset is validated, standardized, and mapped into the locked internal schema before any analysis proceeds.	High	Dataset loads, invalid/corrupted images are rejected, and demographic fields are correctly aligned.

Requirement (User Story)	Details	Priority	Success Criteria
As an auditor, I want to connect a trained model or upload a predictions file, so that I don't need to grant the platform internal access to my model.	Dual-mode Model Interface supports direct PyTorch/TensorFlow inference or batch CSV/JSON predictions ingestion.	High	Predictions are obtained successfully via either mode with no training or weight modification performed.
As a compliance officer, I want a standardized, exportable report, so that I can present it as evidence of EU AI Act Article 10 compliance.	HTML report structured on Model Cards / Datasheets conventions, with every metric traced to its Article 10/Annex IV clause and NIST AI RMF function.	High	Generated report contains a regulatory traceability row for every reported metric.
As a researcher, I want every disparity accompanied by significance testing and confidence intervals, so that I can trust the statistical strength of a finding rather than a bare threshold.	Chi-squared test of independence ($\alpha = 0.05$) and a bootstrap confidence interval ($\geq 1,000$ resamples) reported alongside every Core Four metric.	High	Every non-flagged metric row shows an exact p-value and a 95% confidence interval.
As a non-technical stakeholder, I want plain-language explanations of each metric, so that I can understand what a disparity means without a statistics background.	Report includes accessible descriptions alongside each Core Four metric and explicit caveats on significance and sample size.	Medium	Stakeholder feedback confirms the report is interpretable without additional explanation.
As an auditor, I want flagged disparities explained at the feature level, so that I can tell a genuine demographic effect apart from an unrelated confound.	SHAP-based explainability layer attributes flagged disparities to input features (computed only on flagged disparities, not the full dataset).	Medium	Every flagged disparity in the report is accompanied by a feature-attribution summary.

Functional Requirements

Requirement	Details	Priority	Success Criteria
FR-001 — Data Ingestion	Load, validate, and standardize a benchmark dataset or a compatible custom dataset; check image integrity and align demographic annotation fields (race/ethnicity, perceived gender, age group) into a common internal schema.	High	Corrupted/unreadable files are rejected; every remaining image carries a complete, correctly typed demographic label set.
FR-002 — Dual-Mode Model Interface	Obtain predictions either by direct in-process inference against a supplied	High	Predictions are correctly obtained in

Requirement	Details	Priority	Success Criteria
	PyTorch/TensorFlow model, or by batch ingestion of a precomputed predictions file (CSV/JSON).		both integration modes without model retraining.
FR-003 — Fairness Metric Computation	Compute, per subgroup and per demographic intersection, the four Core Four disparity metrics: DPD, EOD, EOP, DIR using AIF360 and Fairlearn as independent, cross-validating backends.	High	All four metrics are computed by both backends; discrepancies surface as a flagged integration defect.
FR-004 — Statistical Significance Testing	Accompany every reported disparity with a chi-squared test of independence and a bootstrap confidence interval on the subgroup accuracy difference.	High	Every reported metric carries an exact p-value and a 95% confidence interval.
FR-005 — Explainability	Generate SHAP-based feature-attribution output for flagged disparities using a defined strategy: by default, compute SHAP on model embeddings (activations from a chosen intermediate layer) using GradientExplainer/DeepExplainer where supported; raw-image KernelSHAP runs are permitted only with explicit auditor consent and are limited to at most K examples per flagged disparity (configurable; default K=20); every explainability run records the explainer type used and its runtime estimate in the evidence manifest.	Medium	Explainability entries in the report include the explainer type, sample IDs, attribution visualisations, and runtime estimates; raw-image SHAP requires explicit consent and is rate-limited.
FR-006 — Report Generation	Assemble computed metrics, statistical tests, and dataset/model provenance into a standardized, exportable HTML fairness report (Model Cards / Datasheets structure), each row showing sample size, point estimate, 95% CI, and p-value or insufficient-sample flag.	High	Reports correctly and every metric row contains the required fields.
FR-007 — Regulatory Traceability	Map every computed metric to its corresponding clause under Article 10/Annex IV of the EU AI Act and to its NIST AI RMF function.	High	Every metric row in the report links to a specific regulatory clause and RMF function.
FR-008 — Orchestration and Configuration	Expose a command-line configuration interface for selecting the dataset, model/predictions source, and metrics to compute; runs identically given the same configuration and inputs.	Medium	Re-running an identical configuration reproduces identical output.

Requirement	Details	Priority	Success Criteria
FR-009 — Licensing Acknowledgement	Display and require explicit acknowledgement of the selected dataset's licensing terms before an audit is executed.	Medium	An audit cannot start until licensing terms have been explicitly acknowledged.
FR-010 — Regulatory Evidence Traceability	For every audit run and every reported metric, produce a machine-readable evidence manifest (manifest ID, timestamp, code version, dependency lockfile, dataset ID/checksum/licence, model ID and mode, CLI config, random seeds, and computed metrics with sample sizes) and embed it in the exported report bundle as manifest.json.	High	Every exported report bundle contains manifest.json, and its manifest ID is shown at the top level of the HTML report's metadata.
FR-011 — Calibration & Threshold Audits	Support computing per-group calibration curves, expected calibration error (ECE), and thresholded metrics (precision@k, recall at a fixed FPR); allow auditors to request thresholded disparity tables and visualize calibration by subgroup.	Medium	Calibration plots and ECE values are present when requested, and thresholded metrics are included in the export.

Non-Functional Requirements

Requirement	Details	Priority	Performance Metric	Success Criteria
NFR-001 — Statistical Rigor	All significance testing uses $\alpha = 0.05$; exact p-values are reported rather than only a pass/fail flag.	High	$\alpha = 0.05$	Every metric row shows an exact p-value, not a binary flag.
NFR-002 — Uncertainty Quantification	Every reported subgroup accuracy difference includes a 95% confidence interval from a minimum of 1,000 bootstrap resamples.	High	$\geq 1,000$ resamples	CI bounds present on every reported metric.
NFR-003 — Data-Integrity Guard	Any subgroup with sample size below $n = 30$ is automatically flagged as insufficient rather than assigned a computed metric value.	High	$n \geq 30$	No metric is reported for a subgroup below the threshold.
NFR-004 — Performance	Full-dataset evaluation over FairFace's 108,501 images completes within a bounded runtime on a single mid-range GPU; a 5,000-image development subset completes quickly on CPU alone.	Medium	≤ 4 hrs (GPU, full set); ≤ 30 min (CPU, $n=5,000$)	Runtime targets met in test runs.

Requirement	Details	Priority	Performance Metric	Success Criteria
NFR-005 — Modularity	Strict separation of concerns across architectural modules, each with a narrow, well-defined interface, independently unit-testable and extensible.	Medium	1 responsibility / module	Each module passes unit tests in isolation.
NFR-006 — Reproducibility	Bootstrap resampling uses fixed random seeds; third-party dependency versions are pinned in a lock-file to prevent silent behavioural drift.	Medium	Fixed seed + pinned versions	Identical inputs/config reproduce identical results across runs.
NFR-007 — Portability	Runs cross-platform on any environment supporting the pinned Python dependency stack, without proprietary or platform-specific components.	Low	Linux / Windows / macOS	Successful run confirmed on all three target operating systems.
NFR-008 — Explainability Performance	SHAP attribution is computed only on flagged disparities rather than the full dataset by default, to keep explainability overhead a small fraction of total runtime.	Medium	Flagged disparities only	SHAP runtime remains a small fraction of total audit runtime.
NFR-009 — Reproducibility & Report Integrity	Each report bundle includes the code version (git SHA or Docker image digest), a pinned dependency lockfile, the CLI config and random seeds, manifest.json, and an integrity hash (SHA-256) for the report bundle, with an optional HMAC signature if the user supplies a signing key.	High	SHA-256 bundle hash + pinned lockfile	The report bundle includes the lockfile and manifest; the HTML report exposes the code version and bundle hash.
NFR-010 — Statistical Power Guidance	Provide an on-demand power-analysis utility that estimates the minimum detectable effect size given subgroup sample sizes, desired α , and power $(1-\beta)$; for small subgroups, present a sensitivity chart showing the smallest detectable disparity at 80% power.	Medium	<30s runtime (standard inputs)	Power analysis completes in under 30 seconds for standard inputs, with results included in the report as a "sensitivity" panel.

Data Collection: Required Data / Source of the Data

Data Identification

Image data (facial photographs) paired with categorical demographic labels: race/ethnicity (7-level taxonomy), perceived gender (binary), and age group (9 bins), plus true and predicted labels for the audited model's task, and a protected-attribute field selected for disparity analysis.

Data Sources

- FairFace — a face-attribute dataset explicitly built for balanced representation across race, gender, and age; used as the principal validation benchmark.
- UTKFace — previously considered as a secondary benchmark. However, UTKFace is not included in the recommended validation set due to automated DEX age-label noise and racial taxonomy collapse (3/7 categories), preserving statistical integrity.
- Compatible custom datasets supplied by the user, subject to schema validation and licensing acknowledgement.

Data Acquisition Methods

Public benchmark dataset (FairFace) is downloaded from their public repositories. Custom datasets are supplied directly by the auditor. All datasets are validated for image integrity and standardized (resolution, color space via OpenCV) and their demographic annotation fields are aligned into the platform's locked internal schema before any downstream computation proceeds.

Data Collection Challenges

- Licensing compliance for custom, user-supplied datasets, mitigated by requiring explicit licensing acknowledgement before an audit executes (FR-009).
- Small-subgroup statistical insufficiency, any subgroup with $n < 30$ is automatically flagged rather than scored (NFR-003).
- Model-interface incompatibility with non-standard prediction formats are mitigated by the CSV/JSON predictions-file ingestion fallback.

Benchmarking the Model

The platform's own case study benchmarks a FairFace-trained convolutional-network baseline (ResNet-34) evaluated on all four Core Four metrics (DPD, EOD, EOP, DIR), exercising the full ingestion → inference → computation → explainability → reporting pipeline end to end as the platform's validation instance.

Establishing a Baseline Model

A FairFace-trained ResNet-34 CNN serves as the minimum baseline case study against which the platform's fairness-computation and reporting pipeline is validated. Statistical routines (chi-squared and bootstrap resampling) are first unit-tested against known reference values, independent of any real-data run, before this baseline evaluation is performed.

Resources Required

- Hardware (minimum, development): 64-bit CPU, 4+ cores, 8GB RAM, 5GB free disk space.
- Hardware (recommended, full-dataset audit): a single mid-range GPU with ≥8GB VRAM (e.g. NVIDIA T4-class), local or via a free-tier cloud notebook such as Google Colab.

- Software stack: Python 3.10+, PyTorch or TensorFlow, OpenCV, pandas, NumPy, AIF360, Fairlearn, scikit-learn, SciPy, SHAP, Jinja2.
- Cloud services: Google Colab (free tier) for GPU compute, with Google Colab Pro (~\$10/month) as contingency if quota is exhausted; Google Drive for dataset/checkpoint storage (existing free quota).